

Exp No. 8 NLP on clinical Data

Program:

```
!pip install spacy
```

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```

```
https://github.com/explosion/spacy-models/releases/download/en_core_web_sm-3.5.0/en_core_web_sm-3.5.0.tar.gz
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Requirement already satisfied: spacy in /usr/local/lib/python3.10/dist-packages (3.7.5)
Requirement already satisfied: spacy-legacy<3.1.0,>=3.0.11 in /usr/local/lib/python3.10/dist-packages (from spacy) (3.0.12)
Requirement already satisfied: spacy-loggers<2.0.0,>=1.0.0 in /usr/local/lib/python3.10/dist-packages (from spacy) (1.0.5)
Requirement already satisfied: murmurhash<1.1.0,>=0.28.0 in /usr/local/lib/python3.10/dist-packages (from spacy) (1.0.10)
Requirement already satisfied: cytoolz<2.1.0,>=2.0.2 in /usr/local/lib/python3.10/dist-packages (from spacy) (2.0.8)
Requirement already satisfied: preshed<3.1.0,>=3.0.2 in /usr/local/lib/python3.10/dist-packages (from spacy) (3.0.9)
Requirement already satisfied: thinc<8.3.0,>=8.2.2 in /usr/local/lib/python3.10/dist-packages (from spacy) (8.2.5)
Requirement already satisfied: wasabi<1.2.0,>=0.9.1 in /usr/local/lib/python3.10/dist-packages (from spacy) (1.1.3)
Requirement already satisfied: srsly<3.0.0,>=2.4.3 in /usr/local/lib/python3.10/dist-packages (from spacy) (2.4.8)
Requirement already satisfied: catalogue<2.1.0,>=2.0.6 in /usr/local/lib/python3.10/dist-packages (from spacy) (2.0.10)
Requirement already satisfied: weasel<0.5.0,>=0.1.0 in /usr/local/lib/python3.10/dist-packages (from spacy) (0.4.1)
Requirement already satisfied: typer<1.0.0,>=0.3.0 in /usr/local/lib/python3.10/dist-packages (from spacy) (0.12.5)
Requirement already satisfied: tqdm<5.0.0,>=4.38.0 in /usr/local/lib/python3.10/dist-packages (from spacy) (4.66.5)
Requirement already satisfied: requests<3.0.0,>=2.13.0 in /usr/local/lib/python3.10/dist-packages (from spacy) (2.32.3)
Requirement already satisfied: pydantic!=1.8,!=1.8.1,<3.0.0,>=1.7.4 in /usr/local/lib/python3.10/dist-packages (from spacy) (2.9.2)
Requirement already satisfied: Jinja2 in /usr/local/lib/python3.10/dist-packages (from spacy) (3.1.4)
Requirement already satisfied: setuptools in /usr/local/lib/python3.10/dist-packages (from spacy) (75.1.0)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.10/dist-packages (from spacy) (24.1)
Requirement already satisfied: langcodes<4.0.0,>=3.2.0 in /usr/local/lib/python3.10/dist-packages (from spacy) (3.4.1)
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Requirement already satisfied: typing-extensions>=4.6.1 in /usr/local/lib/python3.10/dist-packages (from pydantic!=1.8,!=1.8.1,<3.0.0,>=1.7.4->spacy) (4.12.2)
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.10/dist-packages (from requests<3.0.0,>=2.13.0->spacy) (3.4.0)
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Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.10/dist-packages (from requests<3.0.0,>=2.13.0->spacy) (2.2.3)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.10/dist-packages (from requests<3.0.0,>=2.13.0->spacy) (2024.8.30)
Requirement already satisfied: blis<0.8.0,>=0.7.8 in /usr/local/lib/python3.10/dist-packages (from thinc<8.3.0,>=8.2.2->spacy) (0.7.11)
Requirement already satisfied: confection<1.0.0,>=0.0.1 in /usr/local/lib/python3.10/dist-packages (from thinc<8.3.0,>=8.2.2->spacy) (0.1.5)
Requirement already satisfied: click>=8.0.0 in /usr/local/lib/python3.10/dist-packages (from typer<1.0.0,>=0.3.0->spacy) (8.1.7)
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.10/dist-packages (from typer<1.0.0,>=0.3.0->spacy) (1.5.4)
Requirement already satisfied: rich>=10.11.0 in /usr/local/lib/python3.10/dist-packages (from typer<1.0.0,>=0.3.0->spacy) (13.9.2)
Requirement already satisfied: cloudpathlib<1.0.0,>=0.7.0 in /usr/local/lib/python3.10/dist-packages (from weasel<0.5.0,>=0.1.0->spacy) (0.20.0)
Requirement already satisfied: smart-open<8.0.0,>=5.2.1 in /usr/local/lib/python3.10/dist-packages (from weasel<0.5.0,>=0.1.0->spacy) (7.0.5)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.10/dist-packages (from Jinja2->spacy) (3.0.2)
Requirement already satisfied: marisa-trie>=0.7.7 in /usr/local/lib/python3.10/dist-packages (from language-data>=1.2->langcodes<4.0.0,>=3.2.0->spacy) (1.2.1)
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.10/dist-packages (from rich>=10.11.0->typer<1.0.0,>=0.3.0->spacy) (3.0.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.10/dist-packages (from rich>=10.11.0->typer<1.0.0,>=0.3.0->spacy) (2.18.0)
Requirement already satisfied: wrapt in /usr/local/lib/python3.10/dist-packages (from smart-open<8.0.0,>=5.2.1->weasel<0.5.0,>=0.1.0->spacy) (1.16.0)
Requirement already satisfied: mdurl>=0.1 in /usr/local/lib/python3.10/dist-packages (from markdown-it-py>=2.2.0->rich>=10.11.0->typer<1.0.0,>=0.3.0->spacy) (0.1.2)
✓ 0s completed at 4:45PM
```

```
import spacy
```

```
nlp = spacy.load("en_core_web_sm")
```

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```
# Sample healthcare-related text
```

```
healthcare_text = """
```

The patient was diagnosed with type 2 diabetes and hypertension.

They were prescribed metformin and lisinopril. Further, the patient will undergo

a cardiac stress test at St. Mary's Hospital next week."""

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Process the text

```
doc = nlp(healthcare_text)
```

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```

Print named entities

```
print("Entities in the text:")
```

```
for ent in doc.ents:
```

```
    print(f"{ent.text} - {ent.label_}")
```

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print("Entities in the text:")
for ent in doc.ents:
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```

```
Entities in the text:
2 - CARDINAL
St. Mary's Hospital - FAC
next week - DATE
```

Print explanations of entity labels

```
print("\nEntity Label Explanations:")
```

```
for ent in doc.ents:
```

```
    print(f"{ent.text}: {spacy.explain(ent.label_)}")
```

```
# Print explanations of entity labels
print("\nEntity Label Explanations:")
for ent in doc.ents:
    print(f"{ent.text}: {spacy.explain(ent.label_)}")
```

```
Entity Label Explanations:
2: Numerals that do not fall under another type
St. Mary's Hospital: Buildings, airports, highways, bridges, etc.
next week: Absolute or relative dates or periods
```

Sample healthcare-related text

```
healthcare_text = ""
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```
# Define custom entity labels for diseases and drugs
```

```
disease_labels = ["type 2 diabetes", "hypertension"] # Add more disease terms as needed
```

```
drug_labels = ["metformin", "lisinopril"] # Add more drug names as needed
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# Define custom entity labels for diseases and drugs
disease_labels = ["type 2 diabetes", "hypertension"] # Add more disease terms as needed
drug_labels = ["metformin", "lisinopril"] # Add more drug names as needed
```

```
# Iterate through the text and identify entities based on the custom labels
```

```
entities = []
```

```
for label in disease_labels:
```

```
    if label in healthcare_text:
```

```
        entities.append((label, "DISEASE"))
```

```
for label in drug_labels:
```

```
    if label in healthcare_text:
```

```
        entities.append((label, "DRUG"))
```

```
# Iterate through the text and identify entities based on the custom labels
entities = []
for label in disease_labels:
    if label in healthcare_text:
        entities.append((label, "DISEASE"))
for label in drug_labels:
    if label in healthcare_text:
        entities.append((label, "DRUG"))
```

Print the identified entities

```
print("Entities in the text:")
```

```
for entity, label in entities:
```

```
    print(f"{entity} - {label}")
```

```
# Print the identified entities
print("Entities in the text:")
for entity, label in entities:
    print(f"{entity} - {label}")
```

```
Entities in the text:
type 2 diabetes - DISEASE
hypertension - DISEASE
metformin - DRUG
lisinopril - DRUG
```

Breakdown of the Code:

Loading the Model: We use `en_core_web_sm`, which is a small English language model from spaCy, to process the text.

Processing Text: The healthcare data (e.g., medical conditions, treatments) is fed into the NLP model.

Extracting Named Entities: The program identifies entities like diseases, drugs, organizations (hospitals), and dates, and outputs them with their corresponding labels.

Explaining Entities: The function `spacy.explain()` gives a brief explanation of each entity label (e.g., DISEASE or DRUG).

To create a basic Natural Language Processing (NLP) program for analyzing healthcare data, we can start by implementing a Python program that uses a pre-trained model to perform sentiment analysis or named entity recognition (NER) on health-related texts. Below is an example of a Python program that uses the spaCy library for NER in healthcare data.

Example: Named Entity Recognition on Healthcare Data

en_core_web_sm is a **small** pre-trained language model provided by **spaCy** for English text. It's part of spaCy's core set of models and is designed for general-purpose natural language processing tasks like tokenization, part-of-speech tagging, dependency parsing, named entity recognition (NER), and lemmatization.

Key Features of en_core_web_sm:

1. Small size:

- o The model is lightweight, about 12MB, making it suitable for environments with limited resources.

2. General-purpose:

- o It is trained on a broad dataset (Wikipedia, news articles, etc.), so it works well for a wide range of text but may lack specialized knowledge for certain domains like healthcare, finance, or legal text.

3. Named Entity Recognition (NER):

- o It can recognize general entities such as **PERSON** (people), **ORG** (organizations), **GPE** (geopolitical entities like countries or cities), **DATE**, **TIME**, and more.
- o However, it may struggle with domain-specific entities like **diseases** or **medications**, which is why it's not ideal for healthcare data.

4. Speed:

- o The smaller size means it is faster and more efficient in terms of processing compared to larger models like en_core_web_md or en_core_web_lg.
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